Optimal Confidence Bounds for Sparse Random Projections

Sub-gamma calculus, negative association of moments, and Padé approximants

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🧩 The Johnson–Lindenstrauss Lemma, and Its Constant Problem

The promise. A random $\Pi \in \mathbb{R}^{m \times n}$ with

$$m = O(\varepsilon^{-2} \log(1/\delta))$$

preserves $\|x\|_2$ up to $(1 \pm \varepsilon)$ with probability $1 - \delta$.

Used in nearest-neighbour search, clustering, compressed sensing, sketching for linear algebra, LLM fine-tuning (VeRA).

The catch. That $O(\cdot)$ hides constants that are **orders of magnitude** off.

Concrete symptom

Freksen et al. (2018): the best known constant differs from observed behaviour by a factor ≈ 100.

And the constant enters *quadratically* through ε.

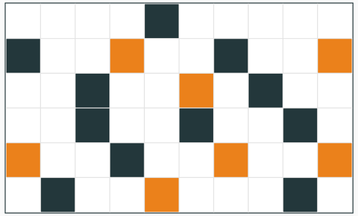
Practitioners must guess m. Theory gives no usable number in the non-asymptotic regime.

Setup: Sparsified Rademacher Projections

$$(\Pi)_{k,i} = s^{-1/2}\eta_{k,i}g_{k,i}$$

- $g_{k,i}$ i.i.d. Rademacher (signs)
- $\eta_{k,i}$ picks s of m rows, *without replacement*, per column (mask)
- $s^{-1/2}$ keeps $\mathbb{E}[\|\Pi x\|_2^2] = \|x\|_2^2$

$s = m$ recovers dense Rademacher. Small $s \Rightarrow$ fast matvec, harder analysis.



n (input) $\times$ m (output), $s = 2$ per column

$+s^{-1/2}$ $-s^{-1/2}$ 0

The object of study: the **distortion** $Q \triangleq \|\Pi x\|^2 - \|x\|_2^2 = x^T(\Pi^T\Pi - I)x.$

🔑 What Would "Optimal" Even Mean?

Lower bounds.

- Jayram–Woodruff, Kane–Meka–Nelson: $m = \Omega(\varepsilon^{-2}\log(1/\delta))$
- **Burr, Gao, Knoll (2018)** — the sharp constant

$$m \geq \frac{4\log(2/\delta)}{\varepsilon^2}(1 + o(1))$$

So 4 is the number to hit — not $O(1)$, not 128.

Three senses of optimality we target

Statistical: match the Gaussian tail $\exp\left(-\frac{\varepsilon^2}{2\,\mathrm{Var}[Q]}\right)$ with *no* multiplicative loss in the exponent.

Algorithmic: attain Burr's leading constant 4 exactly.

Oblivious: stay competitive with dedicated data-oblivious constructions.

Main Result

Theorem 1 — sub-gamma bound for sparse Rademacher projections

Let $v^2 = \mathrm{Var}[Q] = \frac{2}{m}\sum_{i\neq j} x_i^2 x_j^2$ and $c = 3\rho/s$ with $\rho = \|x\|_\infty/\|x\|_2 \le 1$. For $|t| < 1/c$,

$$\mathbb{E}\left[e^{tQ}\right] \le \exp\left(\frac{v^2t^2}{2(1-c|t|)}\right) \quad \Rightarrow \quad \mathbb{P}\{|Q| > \varepsilon\} \le 2\exp\left(\frac{-\varepsilon^2}{2(v^2+c\varepsilon)}\right).$$

Solving for the dimension:

$$\boxed{m \ge \frac{4\log(2/\delta)}{\varepsilon^2}\cdot\frac{1}{1-\frac{6\rho\log(2/\delta)}{s\varepsilon}}} \qquad s > 6\varepsilon^{-1}\rho\log(2/\delta).$$

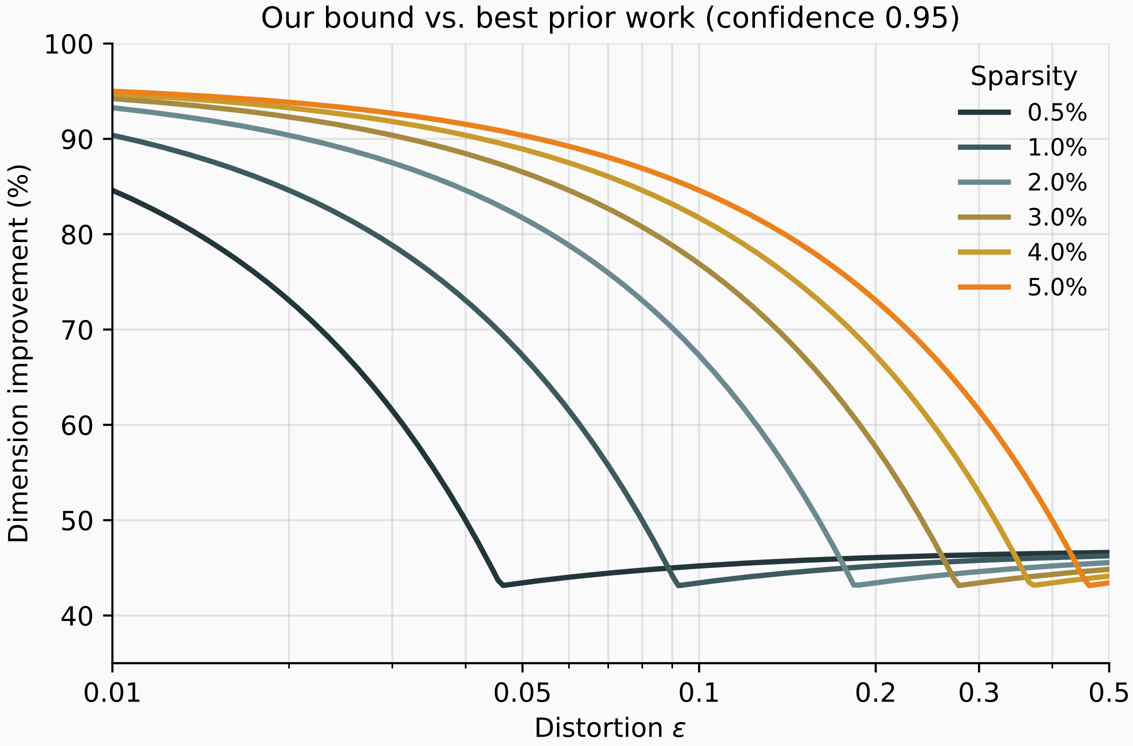
The leading constant is exactly 4. The second factor $\to 1$ whenever $\varepsilon \ll v^2/c$ — the small-to-mid distortion regime that matters in practice.

🔭 Where This Sits in the Literature

Reference	Bound on m	Technique
Necessary conditions		
Jayram–Woodruff; Kane et al.	$\Omega(\log(1/\delta)/\varepsilon^2)$	Communication complexity
Burr et al.	$4\log(2/\delta)(1+o(1))/\varepsilon^2$	Probabilistic argument
Sufficient conditions (prior work)		
Kane–Nelson '10	$\Omega(\max\{\log(1/\delta)/\varepsilon^2, \log^2(1/\delta)/(p\varepsilon)\})$	Rademacher chaos process
Kane–Nelson '12	$\Omega(\max\{\log(1/\delta)/\varepsilon^2, \log(1/\delta)/(p\varepsilon)\})$	Improved moment bounds
Cohen '16	$\Omega(\max\{B\log(1/\delta)/\varepsilon^2, \log_B(1/\delta)/(p\varepsilon)\})$	Matrix Chernoff + majorization
Cohen–Jayram–Nelson '18	$\max\left\{\mathbf{128}\log(1/\delta)/\varepsilon^2, 8\sqrt{2}\log(2/\delta)/(p\varepsilon)\right\}$	Decoupling + MGF
This work		
Theorem 1	$\frac{\mathbf{4}\log(2/\delta)}{\varepsilon^2}\cdot\frac{1}{1-\frac{6\rho\log(2/\delta)}{s\varepsilon}}$	**Sub-gamma calculus**

Prior sufficient conditions are optimal *up to hidden constants*; those constants degrade confidence exponentially.

Quantitative Gain

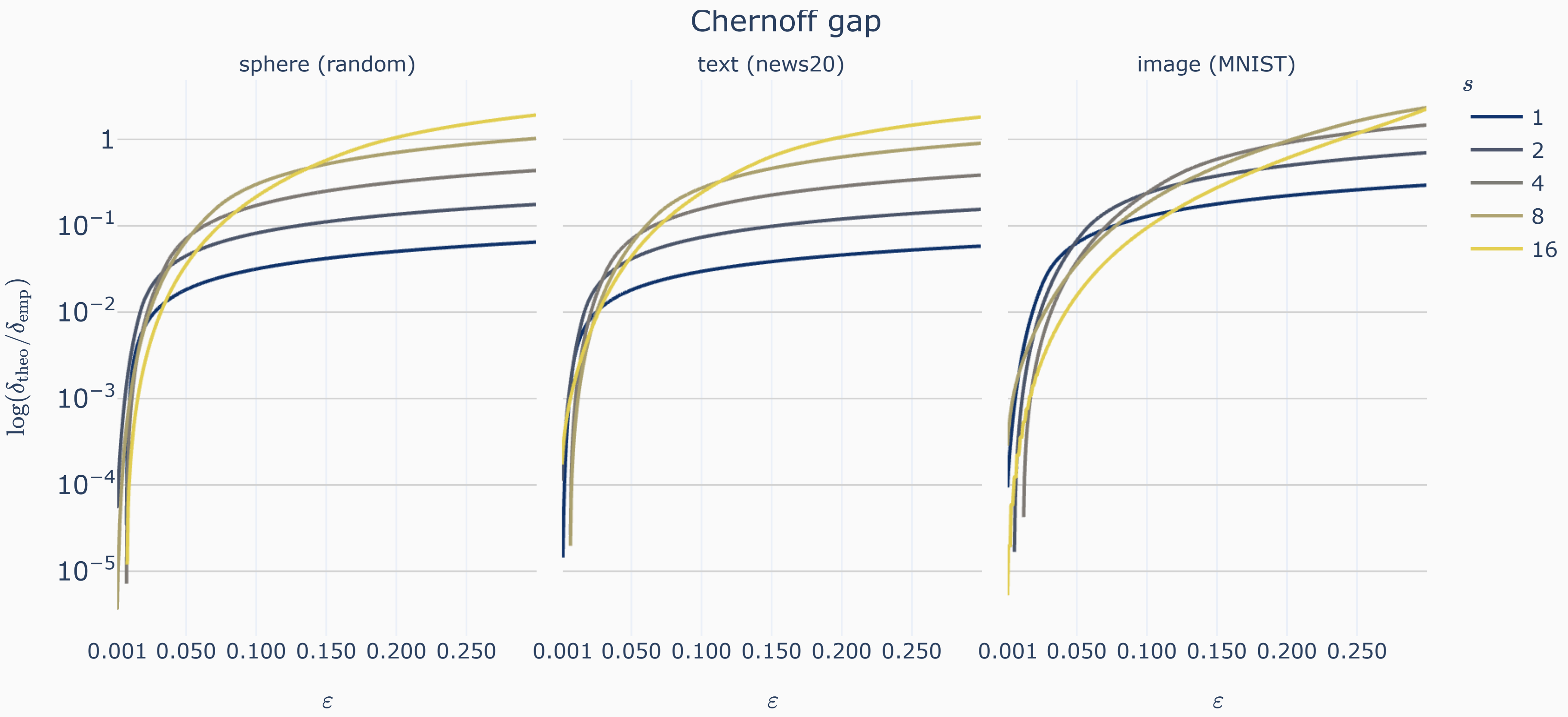


Reading the figure:

- Improvement over the best prior data-oblivious bound at confidence 0.95
- **70–95% dimension reduction** across the practical range $\varepsilon \in [0.01, 0.5]$
- Gains grow with the sparsity budget s

Equivalently: exponentially better confidence at fixed m.

🔬 Is the Bound Actually Tight? Empirical Check



Protocol. Empirical MGF from many projection samples → Chernoff → δ_{emp}, compared against Theorem 1.

For $\varepsilon < 0.05$ the two differ by at most a few percent, and the gap decays to zero with ε — on all three domains, every sparsity.

Proof Roadmap

Decomposition
$Q = \sum_{k=1}^{m} Q_k$

$\downarrow$

Random mask
$Q_k \mid I = s^{-1} g^T X_I g$

$\downarrow$

Eigenvalue form
$Q_k \sim s^{-1} \sum \lambda_i^I (\chi_i^2 - 1)$

$\downarrow$

Eigenvalue bounds
$\|\lambda^I\|_\infty \leq \rho$

$\searrow$

Averaging over mask
Padé approximants

$\downarrow$

Assembly
$\mathbb{E}\left[e^{tQ}\right] \leq e^{v^2 t^2 / (2(1-ct))}$

$\swarrow$

Negative association
$\mathbb{E}\left[\prod Q_k^{d_k}\right] \leq \prod \mathbb{E}\left[Q_k^{d_k}\right]$

$\downarrow$

Tail bound
Chernoff method

Three phases: **(i)** condition on the mask to expose a bounded spectrum; **(ii)** control mixed moments by negative association; **(iii)** assemble a sub-gamma MGF via rational approximants.

🔧 Steps 1–2: Decompose, Then Condition on the Mask

Row decomposition. Since each column has exactly s non-zeros, $\sum_k \operatorname{diag}(\Pi_k)^2 = I$, hence

$$Q = \sum_{k=1}^{m} Q_k, \quad Q_k = x^T \left(\Pi_k^T \Pi_k - \operatorname{diag}(\Pi_k)^2\right) x = s^{-1} \sum_{i \neq j} x_i x_j \eta_{k,i} \eta_{k,j} g_{k,i} g_{k,j}.$$

Masked quadratic form. With $I = \{i : \eta_{k,i} = 1\}$,

$$Q_k \sim s^{-1} g^T X_I g, \quad X_I = \left(x x^T - \operatorname{diag}(x)^2\right)_{I,I}.$$

Spectral form and bounds. Since $\operatorname{Tr}(X_I) = 0$, conditionally on I,

$$Q_k \underset{M}{\prec} s^{-1} \sum_i \lambda_i^I (\chi_i^2 - 1), \quad \left\|\lambda^I\right\|_\infty \leq \left\|\lambda^I\right\|_2 = \left\|X_I\right\|_F \leq \rho, \quad \mathbb{E}\left[\left\|\lambda^I\right\|_2^2\right] = \frac{v^2 s^2}{2m}.$$

Two obstacles remain: the Q_k are *not independent*, and the conditional bound must be averaged over I *without losing constants.*

Tool 1: Negative Association of Moments (NAM)

Definition (NAM)

$X_1, \ldots, X_m$ satisfy NAM if for all non-negative integers α_j,

$$\mathbb{E}\left[\prod_j X_j^{\alpha_j}\right] \le \prod_j \mathbb{E}\left[X_j^{\alpha_j}\right].$$

Equivalent to factorization for all f_j that are power series with non-negative coefficients — **exactly what a Chernoff argument needs**.

Classical NA demands factorization for all *non-decreasing* f_j. A different class, strictly more than we need.

NA genuinely fails here

For $n = 4$, $m = 3$, $s = 2$, $x \propto \left(1, \frac{3}{2}, 2, \frac{5}{2}\right)$, exhaustive enumeration over masks and signs gives

$$\mathbb{E}[f(Q_1)f(Q_2)] > \mathbb{E}[f(Q_1)]\mathbb{E}[f(Q_2)]$$

for $f = \mathbf{1}[\cdot > 0]$.

So NAM is not a convenience — it is the right notion. This settles the open question of Dasgupta, Kumar and Sarlós.

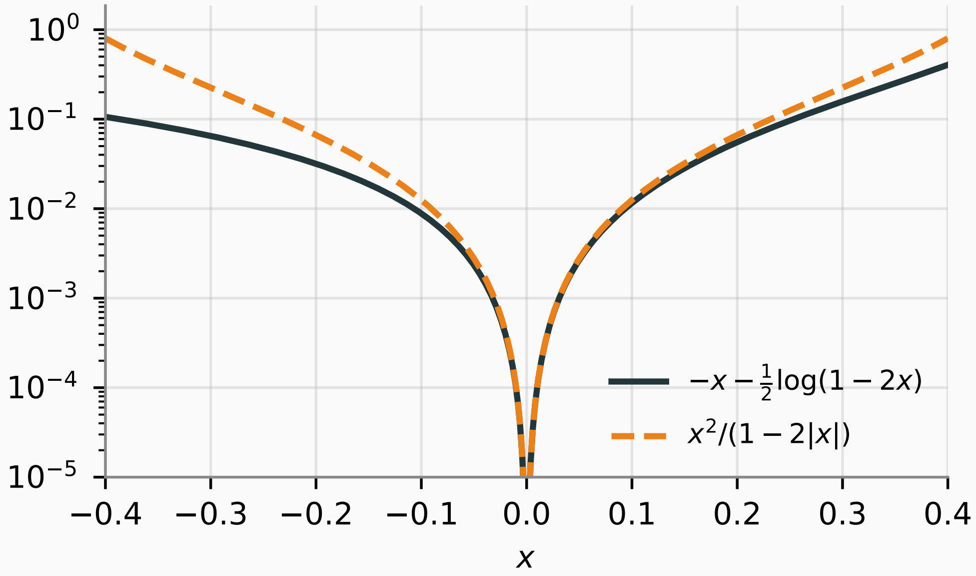
Closure properties. NAM is preserved by (1) multiplication by independent symmetric variables, (2) summation of independent NAM families, (3) summation over partitions. $\Rightarrow$ within-column NA (sampling without replacement) lifts to the row contributions $\{Q_k\}$.

⚙️ Tool 2: Padé Approximants Instead of Taylor

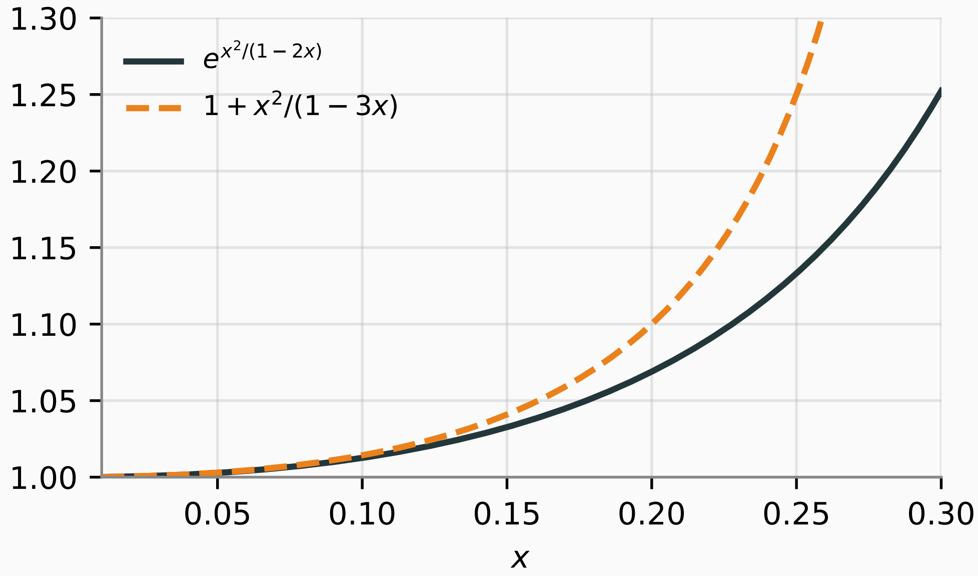
Idea. Rational functions matching the Taylor series to a prescribed order — sharper, over wider intervals, and *linear enough to average.*

Note the sub-gamma form $\frac{t^2 v^2}{2(1-ct)}$ is *itself* a Padé approximant. We nest two of them.

First: bound the log-MGF



Second: make it averageable



The second inequality turns $e^{\cdot}$ into $1 +$ rational $\Rightarrow$ **averaging over I becomes linearity of expectation.**
Both verified by CAD quantifier elimination (Mathematica); both also admit short convexity proofs.

⚙️ Tool 3: Aggregating Conditional Sub-Gamma Bounds

Theorem 3

Let X be a random variable and I auxiliary randomness. Suppose

1. $\log \mathbb{E}[e^{tX} \mid I] \leq \frac{t^2 v_I^2}{2(1-c_I t)}$ *(conditional sub-gamma)*
2. $\mathbb{E}[v_I^2] = \bar{v}^2$ *(variance proxy)*
3. $c_I \leq a v_I$ and $v_I \leq v_{\max}$ a.s. *(variance-scale regime)*

Then for $t < \sqrt{2}/((a+1)v_{\max})$,

$$\log \mathbb{E}[e^{tX}] \leq \frac{t^2 \bar{v}^2}{2\left(1 - (a+1)v_{\max} t/\sqrt{2}\right)}.$$

Why it matters. Naive averaging of conditional bounds loses the variance proxy. Here $\bar{v}^2$ survives intact; only the *scale* degrades, by an explicit factor.

Applied to our setting: $v_I = \frac{\sqrt{2}\|\lambda^I\|_2}{s}$, $c_I = \frac{2\|\lambda^I\|_2}{s}$, $a = \sqrt{2}$, $v_{\max} = \frac{\sqrt{2}\rho}{s} \Rightarrow c = v_{\max}\left(\sqrt{2} + \frac{1}{\sqrt{2}}\right) = 3\rho/s$.

📦 Step 3: Assembly and Tail

Per row (Padé twice, then average over the mask):

$$\log \mathbb{E}\left[e^{tQ_k}\right] \le \frac{v^2 t^2}{2m(1-c|t|)}, \quad c = 3\rho/s.$$

Across rows — NAM lets us multiply MGFs as if the Q_k were independent:

$$\mathbb{E}\left[e^{t\sum_k Q_k}\right] \le \prod_{k=1}^{m} \mathbb{E}\left[e^{tQ_k}\right] \quad \Rightarrow \quad \log \mathbb{E}\left[e^{tQ}\right] \le \frac{v^2 t^2}{2(1-c|t|)}.$$

For $t < 0$: the conditional moments of $Q_k \mid I$ are non-negative (polynomials in symmetric variables), giving the same bound with $|t|$.

Chernoff for sub-gamma variables closes it, with the exact variance

$$\mathrm{Var}[Q] = \frac{2}{m} \sum_{i \ne j} x_i^2 x_j^2 = v^2.$$

No step spends a constant it does not have to.

✨ Spin-off: Hanson–Wright for Randomly Masked Chaos

Theorem 4

Let X be a fixed symmetric $n \times n$ matrix with zero diagonal, I a random index set independent of $g \sim \mathcal{N}(0, I_n)$ (or Rademacher), X_I the principal submatrix on $I \times I$, and $\|X_I\|_F \le B$ a.s. Then for $|t| < 1/(3B)$,

$$\mathbb{E}\left[e^{tg^T X_I g}\right] \le \exp\left(\frac{\mathbb{E}\left[\|X_I\|_F^2\right] t^2}{2(1 - 3B|t|)}\right).$$

Versus standard Hanson–Wright:

- classical versions carry large unspecified constants
- they ignore the masking structure entirely

What we exploit:

- the variance proxy is $\mathbb{E}\left[\|X_I\|_F^2\right]$, not the worst case B^2
- degradation through the mask is controlled explicitly

Substantially improves the recent bounds of Li et al.; useful wherever coordinate subsampling meets quadratic forms.

🥇 Takeaways

Result

- First sparse-JL bound with the **exact** leading constant 4
- Matches Burr's lower bound, not merely up to $O(\cdot)$
- 70–95% dimension reduction over best prior work in practical ranges
- Validated empirically on sphere / text / image inputs

Reusable machinery

- **NAM** — the weakest dependence notion Chernoff needs; NA provably fails here
- **Padé nesting** — sharp bounds that survive averaging
- **Conditional sub-gamma aggregation** — explicit constant degradation
- **Masked Hanson–Wright** — tight constants under subsampling

Practical upshot

`scikit-learn` ships sparse random projections sized by earlier bounds — these results directly shrink the required dimension while strengthening the guarantee.

🚀 Future Directions

1. **Structured matrices.** Circulant and Toeplitz projections for faster embeddings — do the sub-gamma tools survive the loss of row independence?
2. **Data-dependent variants.** Exploit input structure ($\rho = \|x\|_\infty / \|x\|_2$ is already a first step) for adaptive sketching and compressed sensing.
3. **NAM beyond projections.** Which other high-dimensional problems have moment sub-multiplicativity without full negative association?
4. **Beyond the MGF method.** Direct moment methods are in principle slightly better — can they be made to yield comparable constants?

Thank you!

Questions?

Code, notebooks, CAD verification: `https://osf.io/5kr7t`

Backup

🔒 Backup: Verifying the Padé Inequalities

By computer algebra (Cylindrical Algebraic Decomposition, Mathematica):

```
Resolve[ForAll[x, −1/2 < x < 1/2, x^2/(1-2*Abs[x]) >= -x - Log[1-2*x]/2]]
Resolve[ForAll[x, 0 < x < 1/3, Exp[x^2/(1-2*x)] <= 1 + x^2/(1-3*x)]]
```

By hand, both are short.

First: let $f(x) = \frac{x^2}{1-2|x|} + x + \frac{1}{2}\log(1-2x)$. Then $f(0) = f'(0) = 0$ and $f''(x) = \frac{1+2|x|}{(1-2|x|)^3} > 0$, so $f \geq 0$ by convexity.

Second: it suffices that $\frac{x^2}{1-2x} \leq \log\left(1 + \frac{x^2}{1-3x}\right)$. Use $\log(1+s) \geq \frac{s}{1+s}$, then $\frac{x^2}{1-3x+x^2} \geq \frac{x^2}{1-2x}$, which reduces to $x \leq 1$. Both hold on $(0, 1/3)$.

⛓ Backup: NAM Closure Properties

(1) Multiplication by independent symmetric variables. For even α_j, expectations factor across the independent g_j; an odd α_j kills the left-hand side by symmetry.

(2) Summation of independent NAM families. Expand $\left(X_j + Y_j\right)^{\alpha_j}$ by the binomial theorem, apply NAM to each family separately, then recombine — the sums factor back into $\prod_j \mathbb{E}\left[\left(X_j + Y_j\right)^{\alpha_j}\right]$.

(3) Summation over partitions. If $\left\{X_j\right\}_{j \in J_i}$ are independent NAM families over a partition $J_1, \ldots, J_\ell$, then

$$\mathbb{E}\left[\prod_{j=1}^{m} X_j^{\alpha_j}\right] = \prod_{i=1}^{\ell} \mathbb{E}\left[\prod_{j \in J_i} X_j^{\alpha_j}\right] \leq \prod_{i=1}^{\ell} \prod_{j \in J_i} \mathbb{E}\left[X_j^{\alpha_j}\right] = \prod_{j=1}^{m} \mathbb{E}\left[X_j^{\alpha_j}\right].$$

Property (3) is what lifts within-column negative association (sampling s of m rows without replacement, NA by Joag-Dev–Proschan) to the row contributions Q_k.